FortiGate Syslog Integration and Incident Correlation in Wazuh

Employer: Ascent e Digit Solutions Private Limited
Role: Cyber Security Intern
Prepared By: Prakash Raj D
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1. Purpose

This SOP describes the step-by-step procedure to integrate FortiGate firewall logs into Wazuh, validate log ingestion, create custom detection rules, and implement incident correlation for SOC monitoring.

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2. Scope

This procedure applies to:

- Wazuh Manager deployments
- FortiGate firewall syslog forwarding
- SOC monitoring and threat detection use cases

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3. Prerequisites

3.1 Infrastructure Requirements

- Wazuh Manager installed and running
- FortiGate firewall with syslog capability
- Network connectivity between FortiGate and Wazuh server
- UDP port 514 reachable

3.2 Access Requirements

- Root or sudo access on Wazuh server
- Admin access to FortiGate
- Wazuh Dashboard access

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4. Architecture Overview

FortiGate → Syslog (UDP 514) → Wazuh Manager → Decoder → Rules → Alerts → Dashboard

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5. Configuration Steps

5.1 Configure FortiGate Syslog

On FortiGate CLI:

```
config log syslogd setting
   set status enable
   set server <WAZUH_IP>
   set mode udp
   set port 514
end
```

Verify logs are enabled for:

- Traffic
- Event
- System
- VPN

5.2 Configure Wazuh to Receive Syslog

Edit:

/var/ossec/etc/ossec.conf

Add or verify:

```
<remote>
  <connection>syslog</connection>
  <port>514</port>
  <protocol>udp</protocol>
  <allowed-ips>192.168.0.1</allowed-ips>
</remote>
```

Restart Wazuh:

```
sudo systemctl restart wazuh-manager
```

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6. Validation Procedures

6.1 Verify Port Listening

```
sudo ss -lunp | grep 514
```

Expected: UDP 514 listening.

6.2 Verify Network Traffic

```
sudo tcpdump -i any port 514
```

Expected: FortiGate packets arriving.

6.3 Verify Log Ingestion

```
sudo tail -f /var/ossec/logs/archives/archives.log
```

Expected: Raw FortiGate logs visible.

6.4 Verify Alert Generation

```
sudo tail -f /var/ossec/logs/alerts/alerts.log
```

Expected: Parsed Forti alerts.

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7. Custom FortiGate Detection Rules

Create file:

/var/ossec/etc/rules/fortigate_custom_rules.xml

7.1 Base Detection Rule

```
<group name="fortigate,custom,">

  <rule id="110600" level="10">
    <if_sid>81619</if_sid>
    <description>FortiGate suspicious traffic detected</description>
    <mitre>
      <id>T1110</id>
    </mitre>
  </rule>

  <rule id="110601" level="12">
```

```
    <if_group>fortigate</if_group>
    <field name="type">event</field>
    <field name="subtype">system</field>
    <description>FortiGate admin/system activity detected</description>
  </rule>

  <rule id="110602" level="14" timeframe="300" frequency="2">
    <if_matched_sid>110600</if_matched_sid>
    <if_matched_sid>110601</if_matched_sid>
    <description>CRITICAL: Suspicious traffic followed by admin activity</description>
    <group>incident,fortigate,critical,</group>
    <mitre>
      <id>T1068</id>
    </mitre>
  </rule>

</group>
```

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8. Rule Validation

Run:

```
sudo /var/ossec/bin/wazuh-analysisd -t
```

Expected output:

```
Configuration OK
```

Restart manager:

```
sudo systemctl restart wazuh-manager
```

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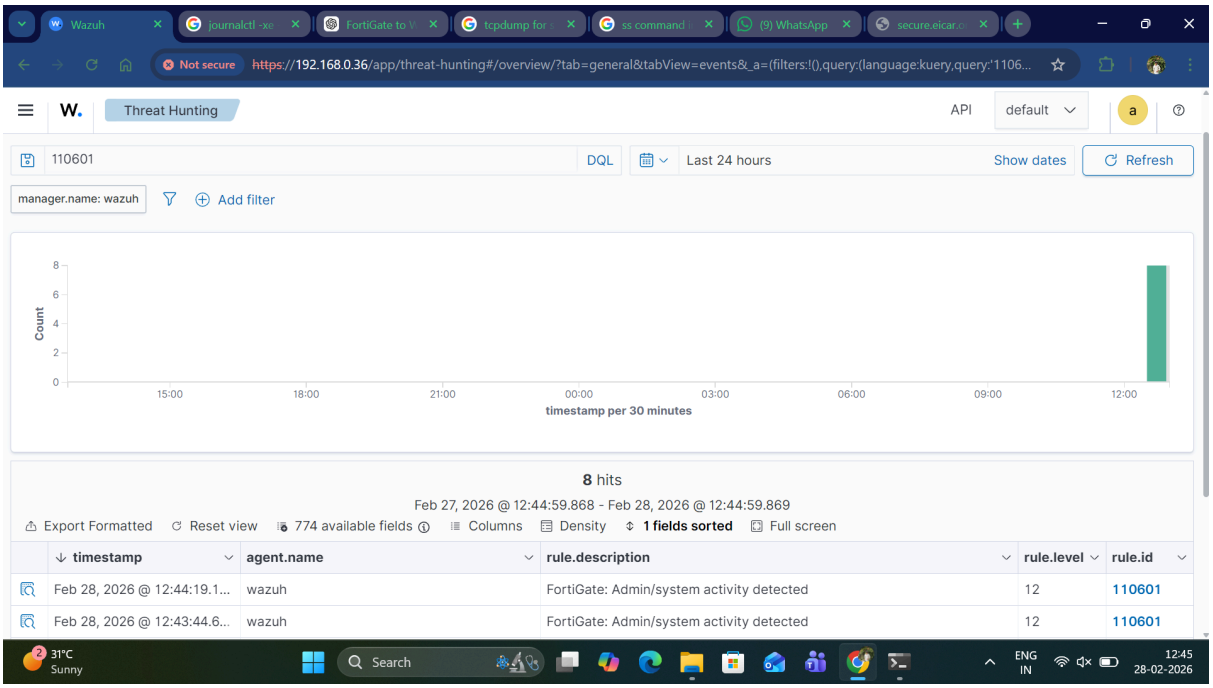
9. Dashboard Verification

Navigate to:

Wazuh Dashboard → Threat Hunting

Test filters:

```
rule.id:110601
rule.id:110602
rule.groups: fortigate
```



Expected:

- Admin/system alerts visible
- Correlated incidents when conditions met

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10. Incident Testing Procedure

10.1 Generate Traffic Burst

From a client machine:

for i in {1..30}; do curl https://google.com; done

10.2 Generate Admin Activity

- Login to FortiGate GUI
- Logout
- Make minor config change

10.3 Expected Result

Wazuh should trigger:

- Rule 110600
- Rule 110601
- Rule 110602 (CRITICAL)

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11. Troubleshooting Guide

Issue: Logs seen in tcpdump but not in Wazuh

Check:

- allowed-ips
- port 514 listening
- firewall rules

Issue: Decoder not matching

Check:

- Forti log format
- Wazuh built-in Forti decoders

Issue: Manager fails to start

Check:

- Duplicate rule IDs
- XML syntax
- Missing group wrapper

Issue: Correlation not firing

Check:

- frequency/timeframe
- both prerequisite rules firing

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12. Security Considerations

- Restrict allowed-ips to firewall IP only
- Use high custom rule ID ranges (110000+)
- Avoid modifying default Wazuh rules
- Validate rules before restart

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13. Maintenance

Daily

- Monitor Security Events
- Review critical alerts

Weekly

- Tune false positives

- Review Forti log volume

Monthly

- Update MITRE mappings
- Review rule effectiveness

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14. Conclusion

This SOP enables:

- Centralized FortiGate monitoring
- Automated threat detection
- SOC-level incident correlation
- Improved security visibility

The implementation is production-ready when all validation steps pass and correlated alerts are observed in the Wazuh dashboard.

End of Document